



AEGIS

A low-cost, modular, **AI-assisted** field network that helps small farmers **save water** and protect crops.

Techathon 3.0 · Vision Venture

Agri-Tech for Sustainable Farming

Team: [Your Name] · [Teammate]

[Your School]

 FARMING RUNS BLIND

Small farmers guess — and pay for it in **water and lost crops.**

~80%

of India's fresh water goes to agriculture — much of it wasted on over-irrigation.

₹ Lakhs

Commercial smart-farming kits cost far beyond a small farmer's reach.

No alert

Frost, heat, dryness and field fires are noticed only after damage is done.

Farmers water on a fixed schedule instead of on need, and get no early warning when conditions turn against the crop.

 THE GAP

Existing systems are
**expensive, fragile and
closed.**

- **Too costly** — priced for large commercial farms, not a 2-acre holding.
- **Not repairable** — one failed part means replacing the whole unit.
- **Needs a smartphone + app** — excludes farmers on basic phones.
- **Dies offline** — no signal, no function, no warning.

 THE IDEA

AEGIS puts a cheap, repairable **sensor node in the field and the **intelligence in the cloud**.**

Each node measures soil moisture, temperature, humidity and smoke. It sends readings over Wi-Fi to the AEGIS cloud, which decides what's happening and replies with plain-language advice — shown on the node's screen, on a live dashboard, and as an **SMS that works on any phone**.

■ HOW IT WORKS

Four layers. The node stays **dumb**; the cloud stays **smart**.

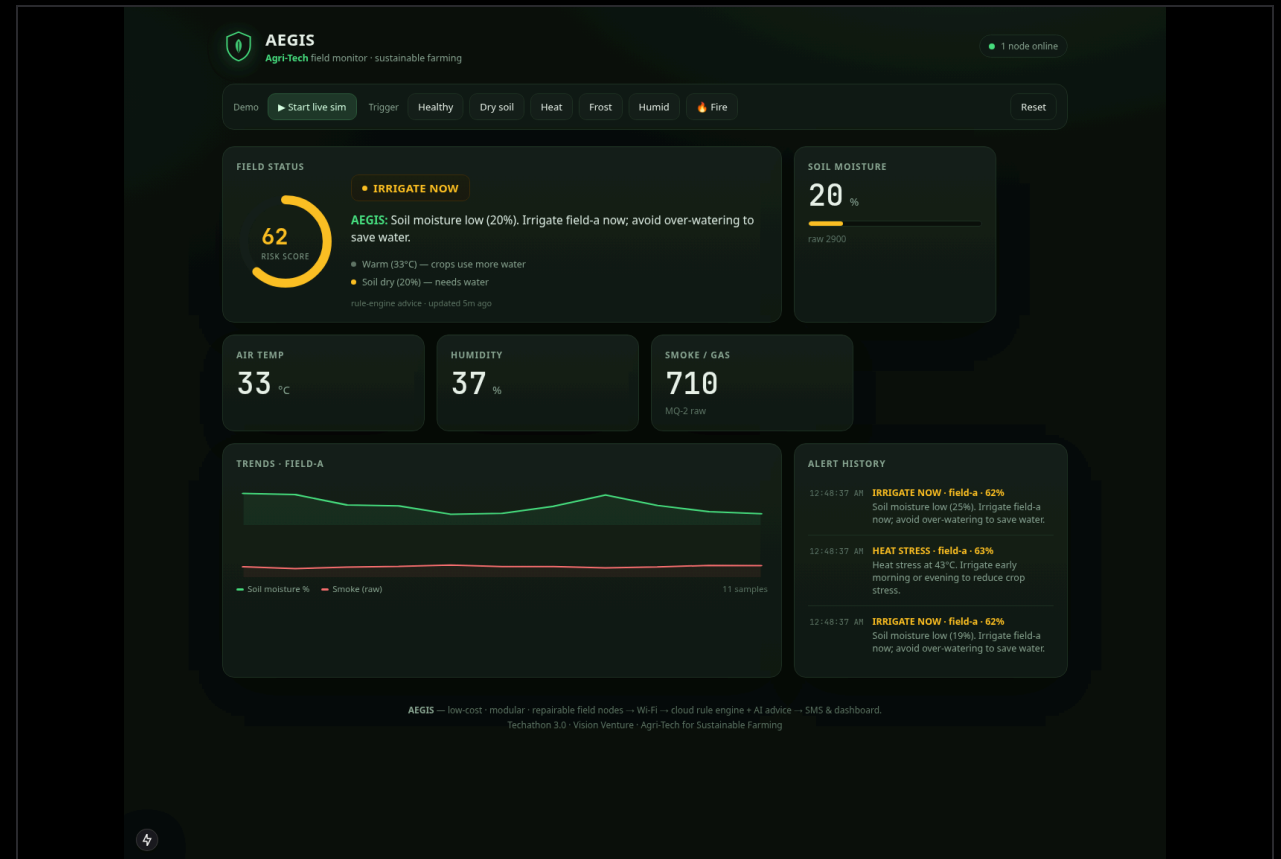


Because the node computes a safe local status too, the farmer **always** sees a warning on the screen — even with no internet. Resilience by design.

WORKING PROTOTYPE

Built, wired, live.

- ESP32 node on a breadboard, reading all four sensors.
- Live cloud dashboard with risk score, trends & alert history.
- Rule engine + AI advice returning real decisions.



AEGIS cloud dashboard — "IRRIGATE NOW", risk 62%, soil 28%.

■ FAST RULES + FRIENDLY AI

Decisions in **milliseconds**, explained like a **human**.

NODE SENDS →

```
{
  "soilPct": 22,
  "temperature": 34,
  "humidity": 38,
  "smoke": 700
}
```

← AEGIS REPLIES

```
{
  "status": "IRRIGATE NOW",
  "risk": 62,
  "message": "Soil low (22%).
  Irrigate field-a now; avoid
  over-watering to save water."
}
```

The rule engine guarantees safety and speed; the AI layer only **rephrases** the facts — so the system still works if the AI is ever offline.

SUSTAINABLE BY DESIGN

Every part of AEGIS pushes toward **less waste**.

Water

Irrigate on *need*, not schedule — directly cuts water use on every field.

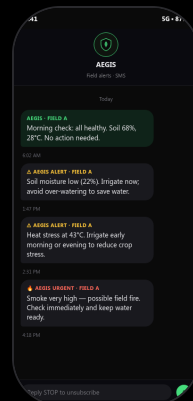
Yield

Early frost / heat / fire / disease warnings prevent crop loss and wasted inputs.

E-waste

Modular, repairable nodes — replace a ₹-cheap sensor, not the whole device.

Accessible: AEGIS sends plain-language **SMS alerts** to any basic phone — no smartphone, no app, no data plan required. The farmer just reads it and acts.



 WHAT MAKES IT WIN

Cheap. Modular.
Unstoppable.

- **~1/100th the cost** of commercial kits — built from commodity parts.
- **Modular PCBs** — each sensor is its own replaceable board.
- **Works offline** — local status on the node when the cloud is down.
- **Any phone** — plain-language SMS, no app needed.
- **Scales** — add nodes across fields; one dashboard tracks them all.

ONE FIELD NODE

A full node for the price of a meal.

COMPONENT	ROLE	APPROX ₹
ESP32 DevKit	controller + Wi-Fi	350
Soil moisture sensor	irrigation signal	60
DHT11	temp + humidity	80
MQ-2	smoke / fire	90
16×2 I²C LCD	on-site readout	150
Wiring, board, misc	—	120
Per node		≈ ₹850

Indicative retail prices; fall further at scale. Cloud runs on free/low-cost tiers.

 WHERE IT GOES

From breadboard to **the field.**

Now ✓

node + cloud + AI + dashboard

**Phase 2**

SMS alerts (SIM/Twilio)

**Phase 3**

solar power · LoRa range

**Phase 4**

auto irrigation valve

**Phase 5**

local-language voice + multi-farm

 THANK YOU

Watch the field. Save the water.

AEGIS — protection that any farmer can afford, repair,
and understand.

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Team: [Your Name] · [Teammate]

[contact / email]